Estimating Implicit Regularization in Deep Learning

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What regularization is implicit in your training procedure?

Why regularization? Explicit regularization adds penalties to the objective function that steer learning toward certain solutions—for example, those with smaller weights—in ways that can reduce overfitting and improve generalization. Training choices such as early stopping, minibatching, and dropout also bias which solutions are learned, but we often don't know exactly how [1, 2].

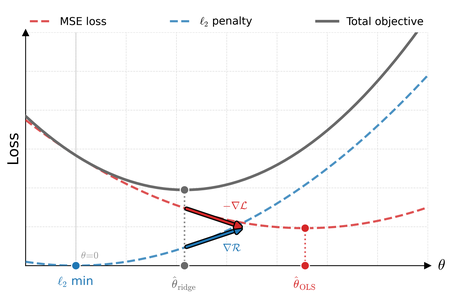
Characterizing implicit regularization. Theory characterizes some of these implicit effects, mostly in simplified settings [3, 4]. We propose an empirical framework for testing theoretical predictions and extending the study of implicit regularization to networks that are not analytically tractable.

Our approach. Work backwards: observe the learned weights and training dynamics, and estimate a parameterized regularizer that explains them. We match loss gradients or update residuals to the gradients of a proposed regularizer and solve for its parameters.

Gradient matching

Training minimizes empirical risk (ERM). We model the learned weights as a regularized ERM (RERM) solution, choosing a family $\mathcal{R}(\theta, D; \Lambda)$ and estimating its parameters Λ:

$$\hat{\theta}_{\text{ERM}} = \arg\min_{\theta} \mathcal{L}(\theta, D) \qquad \longrightarrow \qquad \hat{\theta}_{\text{RERM}} = \arg\min_{\theta} \left\{ \mathcal{L}(\theta, D) + \mathcal{R}(\theta, D; \Lambda) \right\}.$$



An ℓ_2 penalty shifts the solution, where the empirical loss gradient and regularizer gradient cancel.

Estimating from learned weights

At a hypothetical optimum $\hat{\theta}$ of $\mathcal{L} + \mathcal{R}$, the gradients sum to zero:

$$-\nabla_\theta \mathcal{L}(\hat{\theta}, D) = \nabla_\theta \mathcal{R}(\hat{\theta}, D; \Lambda).$$

These $p = \dim(\theta)$ equations let us solve for, or estimate, Λ:

$$\hat{\Lambda} = \arg\min_{\Lambda} \frac{1}{p} \|\nabla_\theta \mathcal{L} + \nabla_\theta \mathcal{R}_\Lambda\|_2^2.$$

Example: For $\mathcal{L} = \|y - X\theta\|^2/n$ and $\mathcal{R} = \lambda\|\theta\|^2$,

$$X^\top(y - X\hat{\theta})/n = \lambda\hat{\theta}.$$

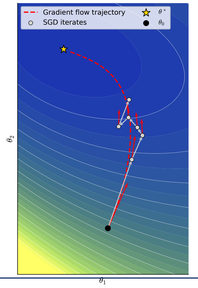
Estimating from weight updates

SGD iterates deviate from the full-batch gradient-flow path (right). For a gradient-step reference, match the update residual:

$$-\frac{\Delta\theta_t}{\eta} - \nabla_\theta \mathcal{L}(\theta_t) \;\approx\; \nabla_\theta \mathcal{R}(\theta_t; \Lambda),$$

where $\Delta\theta_t = \theta_{t+1} - \theta_t$.

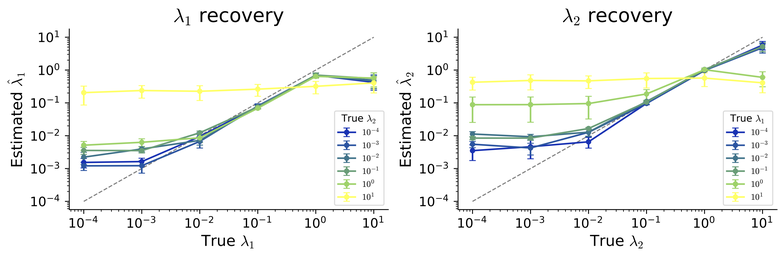
Pool steps to fit a shared regularizer, or let its parameters vary over time. A zero update recovers the endpoint case.



Recovering known regularization

Recover the penalty: match known regularization parameters.
Recover the weights: retrain with the estimated penalty to reproduce the learned weights.

Recovering elastic-net coefficients



Deep ReLU networks; 6×6 penalty grid, 10 datasets per cell. Mean $\pm$ SE; dashed line = exact recovery.

Recovery is accurate at intermediate strengths. Weak signals and nearly collinear gradients limit identification at the extremes.

Early stopping as a quadratic penalty

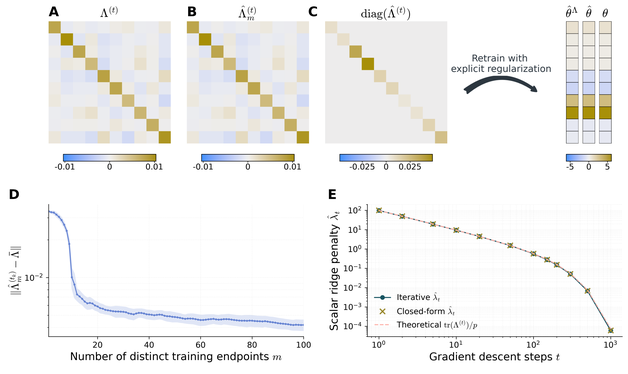
The early-stopped OLS iterate solves a regularized problem [5]:

$$\hat{\theta}_t = \arg\min_{\theta} \left\{ \frac{1}{n}\|y - X\theta\|_2^2 + \mathcal{R}_t(\theta) \right\},$$

$$\mathcal{R}_t(\theta) = \theta^\top \Lambda^{(t)} \theta, \qquad \Lambda^{(t)} = VS\left[(I - \eta S)^{-t} - I\right]^{-1} V^\top.$$

Here $X^\top X/n = VSV^\top$, $\theta_0 = 0$, and $0 < \eta s_i < 1$.

A symmetric Λ has $p(p+1)/2$ unknowns but only p equations per endpoint. For B, we pool endpoints with shared X and t, varying the responses. Full-matrix identification requires diverse endpoints; a diagonal fit has only p unknowns.

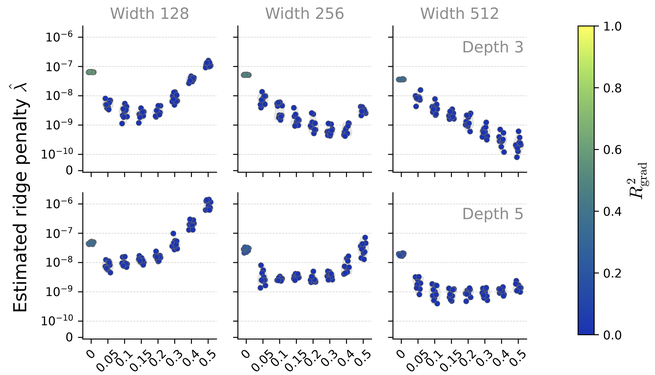


A–B: Theory (A) and recovery from 10 endpoints at the same stopping time (B).
C: A diagonal fit from one endpoint is underparameterized, yet retraining reproduces the early-stopped weights: an equivalent regularizer, despite differing from the theoretical matrix.
D: More endpoints improve matrix estimation, even when stopping times vary.
E: The estimated scalar penalty decreases over training.

Implicit regularization in deep networks

Dropout rate and network architecture

Dropout has data-dependent regularization effects [6]. On MNIST, vary depth, width, and dropout rate; fit $\mathcal{R}(\theta, \lambda) = \lambda\|\theta\|_2^2$.

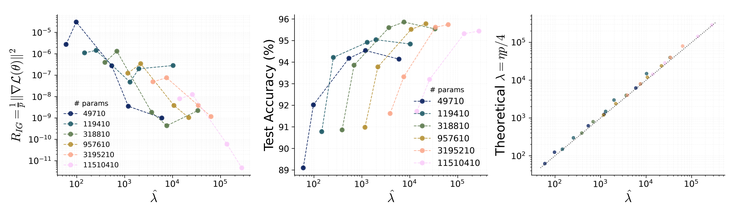


Ten seeds per setting, including zero-dropout controls. Color shows gradient fit: lighter = more explained.

Narrow networks show an upturn at high dropout; wider networks have smaller estimates. Scalar ridge summarizes part of the effect.

Implicit gradient regularization

Finite GD steps approximately follow a modified gradient flow with penalty $\mathcal{R} = \frac{\lambda}{p}\|\nabla\mathcal{L}\|^2$ and strength $\lambda = \eta p/4$ [7]. This penalizes steep parts of the trajectory, biasing learning toward flatter minima associated with better generalization. We recover λ from trajectory deviations (right); stronger regularization accompanies smaller gradients and higher test accuracy (left, middle).



Future work

- **Diagnose training choices early**, before expensive full runs.
- **Study complex architectures and optimizers** through their implicit penalties.
- **Improve identification** with diverse endpoints and richer regularizer families.

References

[1] Bartlett et al. Deep learning: a statistical viewpoint. *Acta Numerica*, 2021.
[2] Smith et al. On the origin of implicit regularization in stochastic gradient descent, 2021. arXiv:2101.12176.
[3] Soudry et al. The implicit bias of gradient descent on separable data, 2018. arXiv:1710.10345.
[4] Arora et al. Implicit regularization in deep matrix factorization, 2019. arXiv:1905.13655.
[5] Ali et al. A continuous-time view of early stopping for least squares regression. In *AISTATS*, 2019.
[6] Wei et al. The implicit and explicit regularization effects of dropout. In *ICML*, 2020.
[7] Barrett and Dherin. Implicit gradient regularization, 2022. arXiv:2009.11162.

Paper: arXiv:2605.05436 — **Blog:** Project exposition — **Code:** GitHub repository

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